

LegalEase AI - Quality Assurance & Testing Guide

Overview

LegalEase AI includes a comprehensive test suite (`tests/`) containing 19 automated test cases covering API routing, schema validation, domain business logic, Gemini AI mocking, sanitization, and multi-format document exporting (TXT, DOCX, and PDF).

1. Running the Automated Test Suite

To run all automated unit and integration tests:

```
powershell  
pytest -v
```

Verified Test Results Summary:

```
text
tests/test_document_generation.py::test_sanitize_text PASSED [ 5%]
tests/test_document_generation.py::test_parse_terms_to_table_data PASSED [ 10%]
tests/test_document_generation.py::test_txt_export PASSED [ 15%]
tests/test_document_generation.py::test_docx_export PASSED [ 21%]
tests/test_document_generation.py::test_pdf_export PASSED [ 26%]
tests/test_document_generation.py::test_html_preview_formatting PASSED [ 31%]
tests/test_generate.py::test_generate_endpoint_success PASSED [ 36%]
tests/test_generate.py::test_generate_endpoint_missing_fields PASSED [ 42%]
tests/test_generate.py::test_generate_endpoint_ai_failure PASSED [ 47%]
tests/test_generate.py::test_generate_endpoint_missing_api_key PASSED [ 52%]
tests/test_health.py::test_root_endpoint PASSED [ 57%]
tests/test_health.py::test_health_endpoint PASSED [ 63%]
tests/test_validation.py::test_valid_document_request PASSED [ 68%]
tests/test_validation.py::test_semicolon_string_terms_parsing PASSED [ 73%]
tests/test_validation.py::test_missing_document_type PASSED [ 78%]
tests/test_validation.py::test_missing_parties PASSED [ 84%]
tests/test_validation.py::test_invalid_date_order PASSED [ 89%]
tests/test_validation.py::test_validate_document_input_helper PASSED [ 94%]
tests/test_validation.py::test_validate_logo_file PASSED [100%]

===== 19 passed, 1 warning in 1.21s =====
```

2. Test Cases Specification

API Gateway Tests (tests/test_health.py & tests/test_generate.py)

Test ID	Test Name	Purpose	Expected Result
TC-API-01	test_root_endpoint	Verifies GET / connectivity	Status 200, message "LegalEase AI API is running"
TC-API-02	test_health_endpoint	Verifies GET /health diagnostics	Status 200, status "healthy" , model & API key flags
TC-API-03	test_generate_endpoint_success	Valid POST /generate payload with mocked AI	Status 200, "success": true , document content returned
TC-API-04	test_generate_endpoint_missing_fields	Payload with missing required fields	Status 422 Unprocessable Entity
TC-API-05	test_generate_endpoint_ai_failure	Simulates upstream Gemini failure	Status 502, friendly error message, no stack trace
TC-API-06	test_generate_endpoint_missing_api_key	Simulates absent GEMINI_API_KEY	Status 500, prompt to configure API key in .env

Validation Tests (tests/test_validation.py)

Test ID	Test Name	Purpose	Expected Result
TC-VAL-01	test_valid_document_request	Standard full payload validation	Passes Pydantic validation cleanly
TC-VAL-02	test_semicolon_string_terms_parsing	Semicolon string parsed into list	List of 3 strings generated
TC-VAL-03	test_missing_document_type	Empty document_type	ValidationError raised
TC-VAL-04	test_missing_parties	Empty parties	ValidationError raised
TC-VAL-05	test_invalid_date_order	end_date set earlier than start_date	ValidationError raised with explicit message
TC-VAL-06	test_validate_logo_file	Corrupted / non-image bytes	Validation fails gracefully

Document Formatting & Export Tests (tests/test_document_generation.py)

Test ID	Test Name	Purpose	Expected Result
TC-DOC-01	test_sanitize_text	Strips markdown fences & normalizes quotes	Clean ASCII text, quotes normalized
TC-DOC-02	test_parse_terms_to_table_data	Parses clauses into table pairs	Generates 4 (Term, Details) tuples
TC-DOC-03	test_txt_export	Exports UTF-8 plain text bytes	Valid bytes, contains title, terms, disclaimer
TC-DOC-04	test_docx_export	Exports Word .docx binary	Valid .docx binary starting with PK\x03\x04
TC-DOC-05	test_pdf_export	Exports ReportLab .pdf binary	Valid .pdf binary starting with %PDF-
TC-DOC-06	test_html_preview_formatting	Renders styled HTML card & table	Contains container div and Schedule A table

3. Manual End-to-End Test Plan

To manually test the complete user experience:

1. Start the backend: `uvicorn backend.main:app --port 8000`
2. Start the frontend: `streamlit run frontend/app.py`
3. In your browser at `http://localhost:8501`:
 - Verify sidebar displays ● **Backend Connected**
 - Select **Employment Contract**
 - Verify default parties and terms
 - Select dates and click **Generate Legal Document**
 - Confirm generated text renders in the dark-themed preview card
 - Click **Click to Edit Document**, modify a term, and click **Save Changes**
 - Click **Download as .TXT**, **Download as .DOCX**, and **Download as .PDF**
 - Open downloaded files and verify formatting, logos, terms table, and footers